

Use Case for a Parking System for

Graduate Certificate

Information Technology – Software Design & Programming

Richard A. Huffman Jr

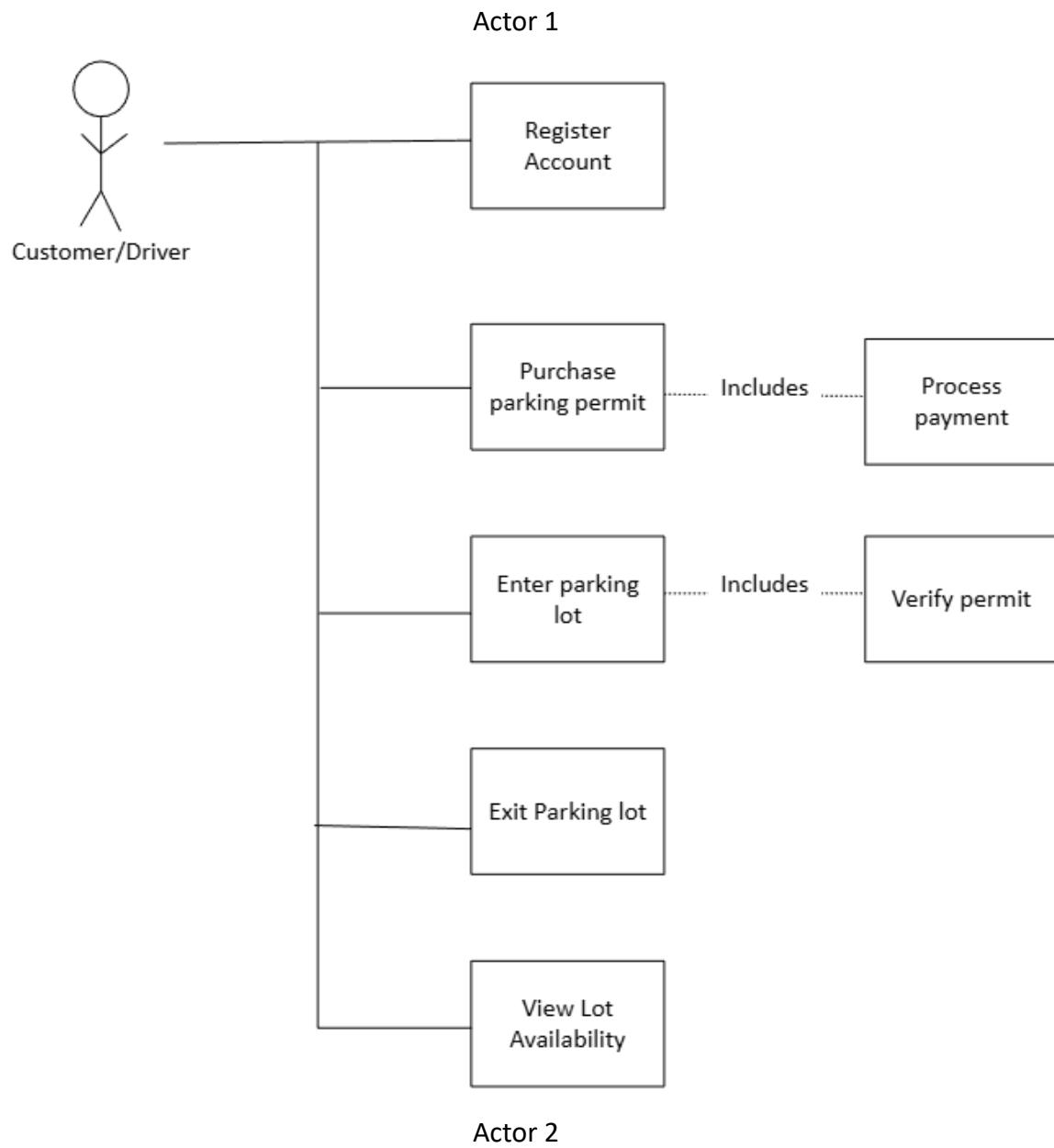
University of Denver College of Professional Studies

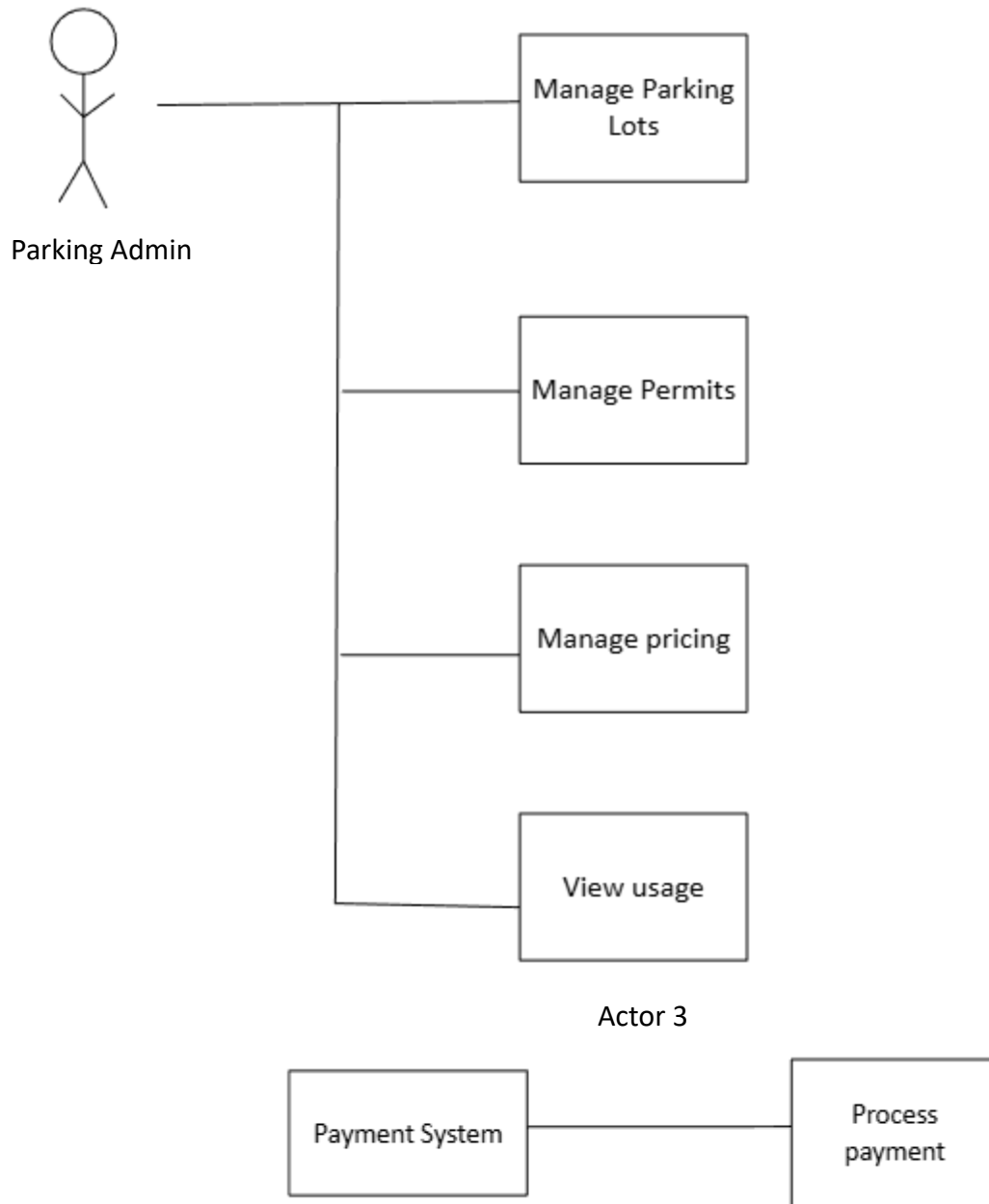
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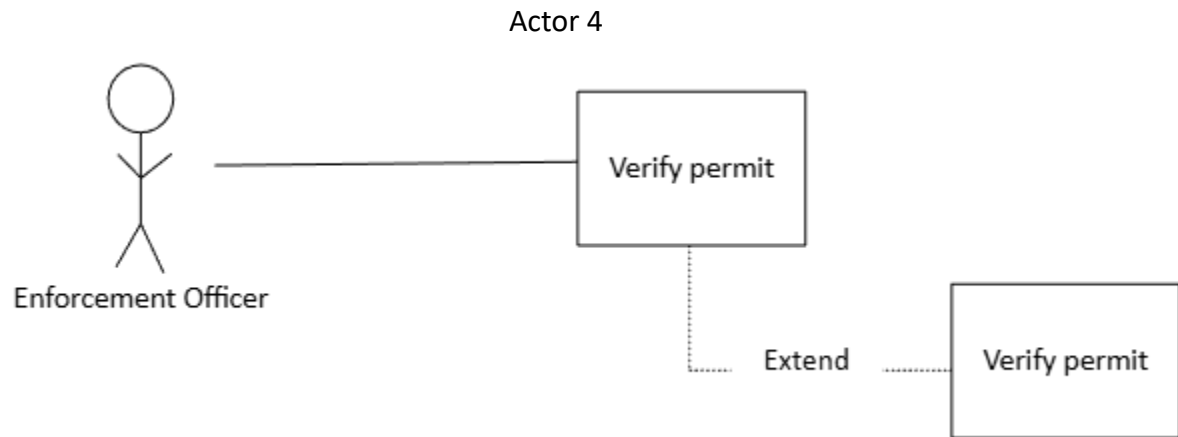
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1. Use Case Diagrams







2. Use Case Elements

Field	Description
ID	UC-01
Name	Purchase parking permit
Short Description	This use case the process a customer takes when purchasing a parking permit through the parking system.
Goal	Allow a customer to obtain a parking permit to park in designated areas.
Preconditions	Operational parking management system The customer can access the system Parking officials/admin have set pricing and permit types Payment processing is operational and available
Success End Condition	Payment processes Permit is created, issued, and active A permit is associated with the customer and the vehicle Customer receives confirmation
Fail End Condition	Payment fails or is declined The permit fails to create The permit fails to associate with the customer or vehicle The system provides an error and does not save changes
Stakeholder	Primary: Customer Secondary: Parking office
Trigger	The customer selects to purchase parking permit on the campus website
Normal Flow	The customer selects a purchase parking permit Logs into the parking management system and selects purchase The system provides a list of permit options and prices The customer selects the desired permit and enters vehicle information The parking management system calculates the total cost and prompts for payment The customer submits payment information The payment system sends information to the payment system The payment system sends confirmation back to the parking management system The parking management system creates and activates a parking permit The parking management system displays a confirmation message to the customer
Alternative Flow	The customer selects a purchase parking permit Logs into the parking management system and selects purchase The system provides a list of permit options and prices The customer selects the desired permit and enters vehicle information The parking management system calculates the total cost and prompts for payment The customer submits payment information The payment system sends information to the payment system The payment system fails to provide confirmation The parking management system displays an error The customer is prompted to retry or cancel No permit is created
Includes	UC-02: Process Payment
Frequency of Use	Multiple times a day, depending on customer demand and permit renewal.
Constraints and Special Requirements	The systems must comply with payment security standards (PCI DSS) Payment processing must happen in real time Permit activation must be immediate
Assumptions	Payment system is reliable and responsive Customer vehicle information is accurate Customer has valid payment information
Notes and Issues	Enhancements may include auto renewal

3. Reflection

The most difficult part of the assignment was avoiding a rabbit hole and providing too much detail or digging too deeply into the technical implementation. For example, in discussing the normal workflow, I could have gone into how payment processing was handled, including the integration with third-party processors. Another challenge was deciding on the use cases and actors. Did I want to have a system that automatically determined how many spaces were available? Ultimately, the number of use cases could grow rapidly and put me out of scope for the system's intended purpose.

What helped with the assignment was taking a step back and re-reading the instructions. The instructions clearly state what the intent is and what needs to be accomplished. The examples gave me a better understanding of what I need to do to complete the assignment. I wish I had read the instructions more closely before I started the assignment, so I would have known what to do. A consequence of not knowing was sitting there, trying to figure out how to capture everything, and being overwhelmed in the process.

Several implementation decisions were made to keep the system more manageable. One example is that the parking management system and the payment system are separate. That method, to me, reflects a real-world system where the application you are using to make the purchase is not necessarily the one that handles payment processing. For me, this assignment highlights the usefulness of developing use cases as they provide a clear requirement for future development efforts.